The consensus model is a hotly debated topic in blockchain technology. Participants enabled by the self-assigned identity can send transactions, and these transactions are collected into different blocks by the blockchain (miner) nodes, as shown in figure 12.3. One of the many blocks formed will be appended to the chain. The challenge is to add a block about which the stakeholders in the network agree. Let’s explore this aspect next.

Consensus means agreement among peers. This consensus is an agreement among the full nodes about the next block to be added to the chain, a process that ensures the integrity of the chain. To address this issue, different consensus models--such as proof of work (POW), proof of authority (POA), and proof of stake (POS)--have been proposed and tried.

Bitcoin uses proof of work (POW) for consensus. Figure 12.3 shows a high-level depiction of POW consensus. The figure shows that miners compete (by solving a computation puzzle) to add the next block to the chain. POW is computationally intensive and results in enormous power consumption in the massive racks of specialized computers used in solving the POW puzzle for the right to mine the next block. It is estimated that Bitcoin mining consumes as much as energy as the country of Ireland uses per day. Therein lies the problem. Let’s discuss the POW method and two alternatives to it.